

Model comparison — every family tried, regime and verdict

temporal-exploit · 2026-07-03 · numbers read live from [artifacts/](#) at build time; results without a persisted artifact are cited from committed docs and marked **doc** in the Source column.

How to read this

Two targets. (1) **In-wild** — time to known-exploited catalog evidence (KEV / VulnCheck; Google 0-day adds 0 usable events — all pre-publication); rare (~396 events on handover labels, 1,310 with VulnCheck merge). Honest framing: ~93% of merged labels are VulnCheck catalog-add dates — a publication-anchored known-exploited *proxy*, not exploitation onset. (2) **First-weaponization** — earliest public exploit signal; abundant (45.5k test events) but ~97% PoC dates, so it measures time to public exploit *tooling*. **Two regimes.** A locked 2024-01-01 time split (c-index/IPCW) for at-scale comparisons, and the decisive **quarterly rolling-origin backtest** (15 scheduled origins; runs score 14 when a low-event origin is skipped; train only on what was knowable at each origin) for anything claiming deployment value. Backtest beats split when they disagree — several single-split 'wins' below died prospectively. **Numbers are not comparable across tables:** Table A is the 396-event handover cohort, Table B the 1,310-event merged cohort with a different model and feature set — Cox 0.817 (A) vs structural-XGB 0.816 (B) at AUC@90 is coincidence, not a tie.

A. In-wild target — survival model families (rolling-origin backtest)

Model	Headline (backtest)	Data	Source	Verdict
Penalized Cox PH (+EPSS feats)	AUC@90 0.817±0.069 · recall@top-10% 0.51	396 ev / 14 origins	inwild_headtohead.json	In-wild backbone. Wins every axis that matters (discrimination, recall, variance, speed; GBM edges the IPA point estimate, statistically tied); IPA=0 is the rare-event reality. Config predates the 2026-07-02 EPSS-free directive (which governs Table B's comparison arms, not this family grid)
Random Survival Forest	AUC@90 0.770±0.117 · recall 0.42	396 ev / 14 origins	doc: progress.md \$head-to-head (artifact not persisted)	Loses to Cox — matches Burk et al. 2026 (nothing beats Cox on tabular survival)
Gradient-boosted (sksurv Cox loss)	AUC@90 0.710±0.093 · recall 0.40 · Δvs cox -0.107 [-0.154,-0.061]	396 ev / 14 origins	inwild_headtohead.json	Loses; O(n²) loss forces 4k-row subsample that discards scarce events (documented structural handicap)
Mixture cure (log-logistic latency)	single-split IPA +0.003 → backtest negative; recalibrated variant worse (IPA@180 -0.27)	396 ev	doc: README/progress (backtest overturned the split win)	Dead-end. Non-identifiable — no KM plateau; ~99.5% censoring is administrative, not a cured fraction. Do not revive
Stacked transfer (first-weap Cox risk as covariate)	Δ=0 vs Cox	396 ev	doc: research_pathways_2026-06.md	No gain — the source model shares the target's features
XGBoost-AFT (structural, GPU)	AUC@30 0.693 · AUC@90 0.724 (2026-07-03 refreshed 81-col artifacts)	396 ev / 14 origins	doc: improvement_log_2026-07-02.md \$Final state	Loses to Cox at rare-event scale — the scale flip vs Table C is the story
Temporal recalibration (Booth baseline refit)	ΔAUC@90 +0.0000 (rank-preserving) · ΔIPA@90 mean -0.0008 / median +0.0003 over 15 origins	1,310 ev / 15 origins	inwild_temporal_recal.json	OFF — a wash, not a win: the clock-floor fix already removed the staleness bias; refitting on a smaller recent window only adds variance
Temperature recalibration (cross-fit S^exp(a))	AUC unchanged (provably rank-preserving) · IPA@90 -0.003 → -0.092	396 ev	doc: progress/README 2026-06-19 (no JSON artifact)	OFF — event-starved origins learn bad temperatures; harmful on the mean

B. In-wild target — vs the EPSS baseline (the corrected comparison)

Arm	Numbers (15-origin backtest)	Source	Verdict
Raw EPSS percentile (the honest baseline)	AUC@30 0.695 · AUC@90 0.683 · PR-AUC@30 0.018 · recall@top-1% 0.068	inwild_epss_parity.json	The bar. Rank by the raw score — never wrap it in a model
XGB-AFT trained ON EPSS features (naive wrap)	AUC@30 0.501	inwild_epss_parity.json	Collapses to chance — AFT loss at 99.5% censoring can't exploit a probability-shaped feature. Measurement-bug class: a wrapped baseline inflates any 'we beat EPSS' claim

Arm	Numbers (15-origin backtest)	Source	Verdict
Structural XGB-AFT (NO EPSS data — standing directive)	AUC@30 0.795 · AUC@90 0.816 · Δvs raw EPSS +0.100 [+0.055,+0.145] @30, +0.134 [+0.089,+0.178] @90 · PR-AUC@30 Δ+0.005 (CI crosses 0)	inwild_epss_parity.json	Headline. Beats raw EPSS on ranking; PR-AUC statistically tied at ~1310 positives; EPSS still wins recall@top-1%
+ publication-EPSS as features (ablation)	full-no_epss AUC@30 -0.114 [-0.201,-0.027]	inwild_epss_ablation.json	EPSS features HURT the xgb model — supports the EPSS-free directive empirically
+ landmark-EPSS trajectory (L=30 ablation)	full-no_epss AUC@30 +0.034 [-0.056,+0.124] · AUC@90 -0.043 [-0.088,+0.003]	inwild_epss_ablation_landmark.json	Statistical wash — even the EPSS trajectory adds nothing over structural for ranking

C. First-weaponization at scale (locked 2024-01-01 split; 212,824 train / 101,023 test, 45,527 test events)

Model	Discrimination	Calibration	Source	Verdict
XGBoost-AFT (GPU)	IPCW c-index 0.614 · mean AUC(t) 0.626	integrated Brier 0.240	report_fw_headtohead/metrics.json	Discrimination headline at scale; calibration mediocre (slope ~0.2, doc D2)
Cox PH (penalized)	IPCW c-index 0.562 · mean AUC(t) 0.576	integrated Brier 0.220	report_fw_headtohead/metrics.json	Calibration winner (slope ~0.6) + interpretable reference; 23/69 PH violations per the 2026-06-12 enriched run (doc — the persisted PH artifacts return NaN p-values, 0 flagged)
DeepSurv (pycox, CUDA)	c-index(td) 0.575	integrated Brier 0.287 (20k-row eval)	report_fw_headtohead/metrics.json	Wins neither axis — deep survival does not pay at this scale

Fine print: c-index values are IPCW-corrected on the full test set; DeepSurv's time-dependent concordance is computed on a 20k-row evaluation subset (memory budget). 'Discrimination headline' is specific to IPCW c-index/AUC(t): the same artifact's paired truncated-C bootstrap favors Cox (Δ=0.009, CI excl. 0) — metric-dependent, both reported. The framing caveat applies to this whole table: ~97% of events are PoC dates.

D. Competing risks & deep discrete-time

Model	Headline	Source	Verdict
Cause-specific Cox (exploitdb cause)	test c-index 0.728 (3,794 events)	report_competing_deep/competing_metrics.json	Working competing-risks model; only the rarest cause (google_0day, 1 event) is unscoreable — other rare causes score on thin test-event counts
Aalen-Johansen CIF vs independent KM (inflation @30d)	independent-KM inflation vs AJ: poc 1.3% · other causes 3.0–6.8%	report_competing_deep/competing_metrics.json	Dependence smallest for the dominant PoC cause, larger for rare causes on this split; 13–18% relative at 180d on the merged corpus (doc: pipeline_characterization_2026-06.md §3)
DeepHit (as stored, pre-fix: equal-width bins)	poc CIF@90 3.0e-06 (AJ truth ~0.10) · c-index(td) 0.57; rare causes ~1e-11 at ~0.50	report_competing_deep/competing_metrics.json	Collapsed on rare-event imbalance
DeepHit (quantile time-bins fix)	CIF@90 3e-6 → ~0.17 (AJ truth ~0.10)	doc: deephit_imbalance_fix_2026-06.md (artifact predates fix)	Fixed and usable — still does not beat Cox (data-limited)
PoC→KEV transition head (clock restarts at PoC)	test c-index 0.869 [0.839, 0.897] (single split + bootstrap; 83 test events) · contrast: PoC→MSF 0.53 near-chance, PoC→Nuclei 0.79	doc: pipeline_characterization_2026-06.md	Best discrimination in the repo; deployable as the STATE-2 escalation (recall@top-10% 0.50, ~4d median lead — defender_interpretation doc)

E. Config & operating-point comparisons

Variant	Numbers	Source	Verdict
Feature hill-climb (gated greedy, leakage-safe groups)	EPSS+CVSS beats EPSS-only: recall@top-10%@30d 0.103→0.130, Δ+0.027 CI[0.013,0.042]; then plateau	doc: beat_epss_attempt_2026-06.md	Only CVSS adds signal over EPSS; confirms the data-limited ceiling
Operating points (wait-L landmark heads)	PR-AUC@30: L=0 0.0071 → L=7 0.0376 → L=30 0.0460 · median lead 144→226d	operating_points.json	Waiting 30d buys ~6× PR-AUC@30 — the wait-vs-lead-time trade IS the deployment message

Variant	Numbers	Source	Verdict
Era stress (non-stationarity)	2022→2024 degradation -0.031; 2023→2025 +0.074	era_stress.json	Era-dependent, opposite signs — inconclusive; retrain per origin regardless
Clock-start floor (KEV catalog launch; xgb, merged labels)	floored AUC@90 0.811 (1201 ev) vs unfloored 0.829 (1310 ev)	inwild_floor_ablation.json	Backfill-artifact guard changes little; kept for honesty
XGB early stopping (random event-stratified val)	25% faster (32.8→24.5 s) but AUC@30 -0.043 [-0.081,-0.006], AUC@90 -0.043 [-0.066,-0.021]	xgb_earlystop_ab.json	Adoption rejected 2026-07-03 — tens of val events/origin make aft-nloglik noisy; ships opt-in

The standing verdicts

1. Penalized Cox is the in-wild backbone; structural XGB-AFT (EPSS-free) is the EPSS-comparison headline (+0.100 AUC@30 over raw EPSS; PR-AUC indistinguishable — underpowered at ~1,310 positives, not a demonstrated tie). **2.** XGB-AFT is the discrimination headline at first-weaponization scale; Cox wins calibration. **3.** Documented dead-ends — do not revive without new evidence: mixture cure (non-identifiable), RSF/GBM (lose on in-wild), DeepSurv/DeepHit (lose at first-weap scale; not viable at ~400 events per the EPV literature), stacked transfer ($\Delta=0$), recalibration on in-wild (event-starved), EPSS-as-feature (hurts), EPSS-wrapped baseline (collapses — measurement bug), shared-frailty illness-death (NO-GO 2026-06-20: ~2,577 transition events + R-outside-harness). SurvivalBoost is wired ([boost]`extra) but never benchmarked — the one untried family. **4.** The in-wild ceiling is data-limited (396 handover events; 1,310 merged — the extra VulnCheck labels tighten CIs via n, not sharper timing), not model-limited — proven by exhausting the model space above. **5.** Next levers (specced 2026-07-03): re-metric to coverage/effort + recall@K with CIs; LambdaRank top-push; multi-seed + bounded tuning; Vulnrichment SSVC git-history labels — the one lever statistics says can move PR-AUC. EPSS-version caveat: all numbers above are vs EPSS v3/v4 history; EPSS v5 shipped 2026-06-15.

Build: scripts/build_model_comparison_report_2026-07-03.py · sources: artifacts/*.json + committed docs as labelled · pipeline state: speed-memory-bundle merged 2026-07-03 (record: docs/improvement_log_2026-07-02.md).